Mid-Sem Exam

612203012

Question 1

```r
x = c(1, 2, 3, 4, 5)
f = c(7, 11, 9, 8, 3)
d = rep(x, f)
#a) Mean
mean(d)
```

```
## [1] 2.710526
```

```r
#b) Median
median(d)
```

```
## [1] 3
```

```r
#c) Mode
t = table(d)
m = names(t)[which(t > 1 & t == max(t))]
m
```

```
## [1] "2"
```

```r
#d) Seventh decile
quantile(d, 0.7)
```

```
## 70%
##   3
```

```r
#e) 29th percentile
quantile(d, 0.29)
```

```
## 29%
##   2
```

Question 2

```r
price = c(10, 15, 30, 42, 50, 60)
qty = c(4, 20, 15, 10, 16, 8)
df = data.frame(price, qty)
df #Required dataframe
```

```
##   price qty
## 1    10   4
## 2    15  20
## 3    30  15
## 4    42  10
## 5    50  16
## 6    60   8
```

```r
value = price * qty
df = cbind(df, value)
df #df with vector value = price*qty added
```

```
##   price qty value
## 1    10   4    40
## 2    15  20   300
## 3    30  15   450
## 4    42  10   420
## 5    50  16   800
## 6    60   8   480
```

Question 3

```r
age = c(22, 27, 31, 41, 30, 25, 19, 20, 23, 35)
#a) Ages of persons except 5th and 7th
age[c(-5, -7)]
```

```
## [1] 22 27 31 41 25 20 23 35
```

```r
#b) Vector containing ages of persons greater than 30
age30 = age[age > 30]
age30
```

```
## [1] 31 41 35
```

```r
#c) Ages of 4th, 5th and 6th persons
age[4:6]
```

```
## [1] 41 30 25
```

```r
#d) Vector containing ages of persons between 20 and 25
age2 = age[age >= 20 & age <= 25]
age2
```

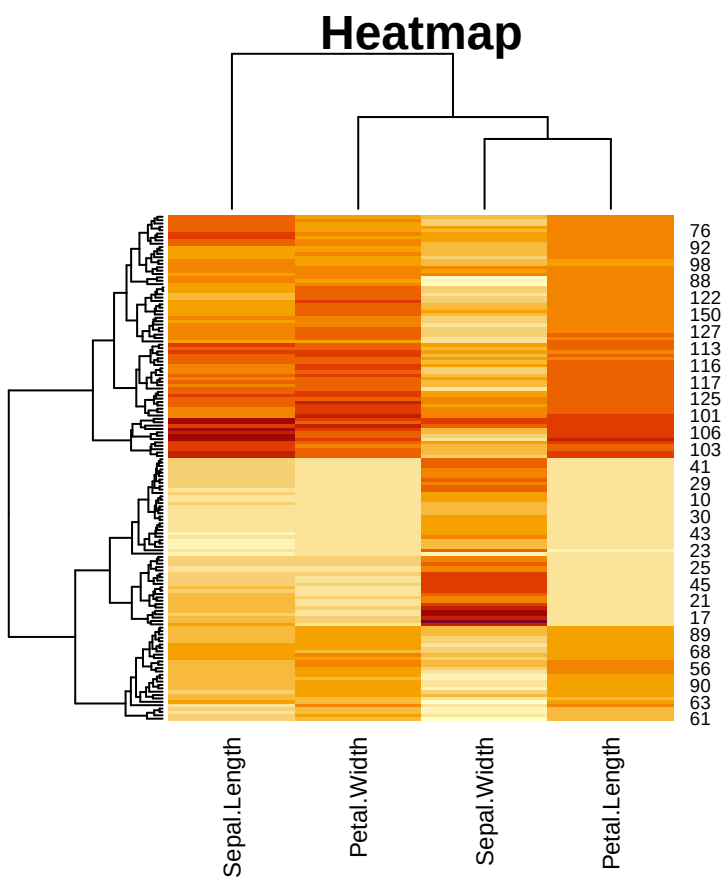
```
## [1] 22 25 20 23
```

```r
#e) Ages of last 3 persons
age[8 : 10]
```

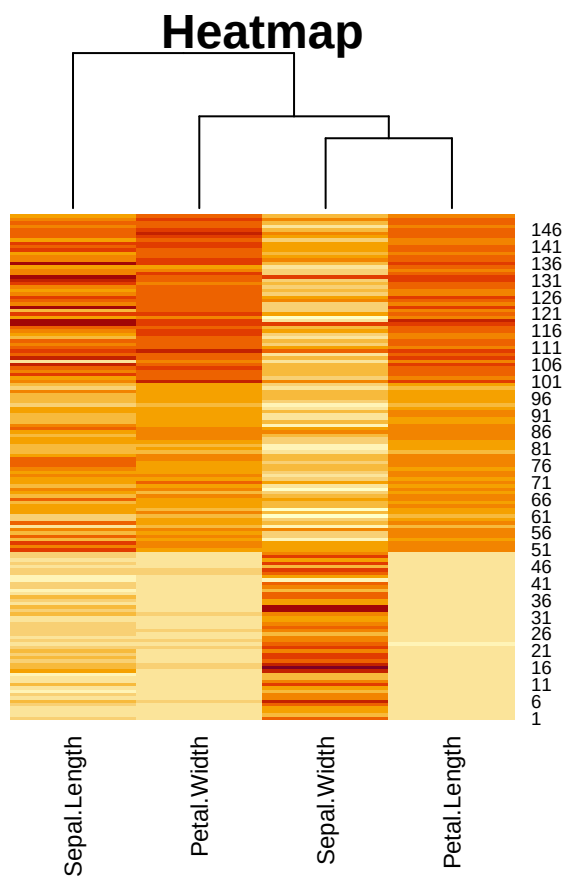
```
## [1] 20 23 35
```

Question 4

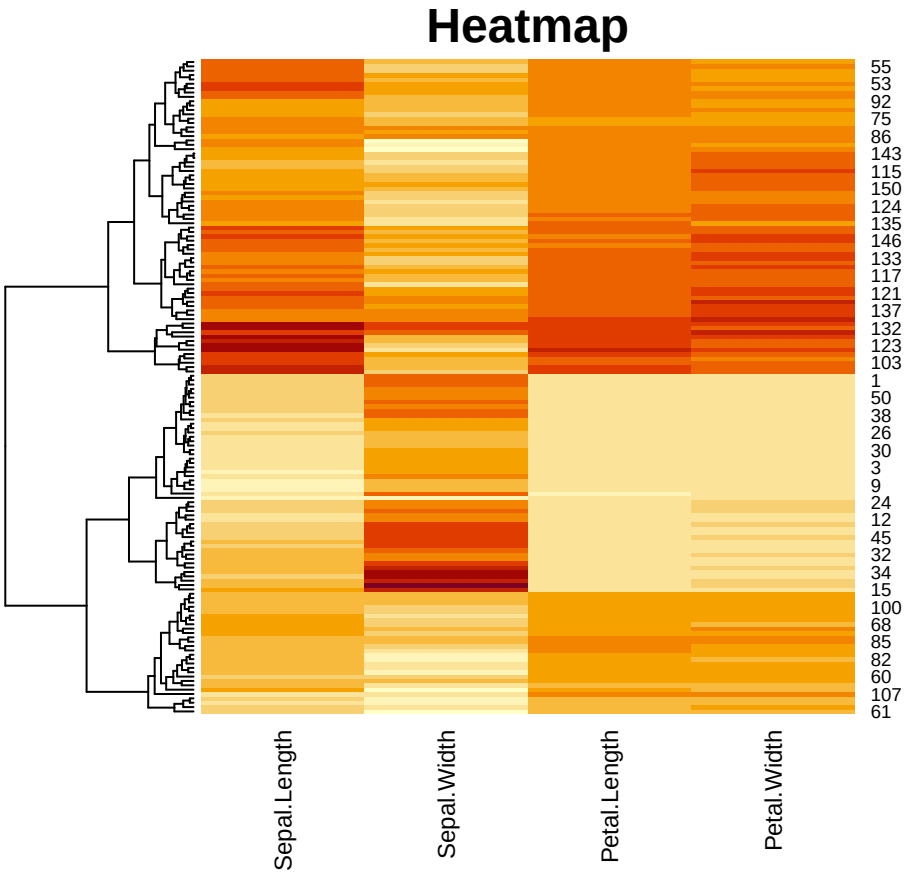
```
data(iris)
d = iris[-5]
heatmap(as.matrix(d), main = "Heatmap", scale = "column", cexCol =
↪  0.9)
```




```
#a) Heatmap without row dendrogram
heatmap(as.matrix(d), main = "Heatmap", scale = "column", Rowv = NA,
↪  cexCol = 0.9)
```

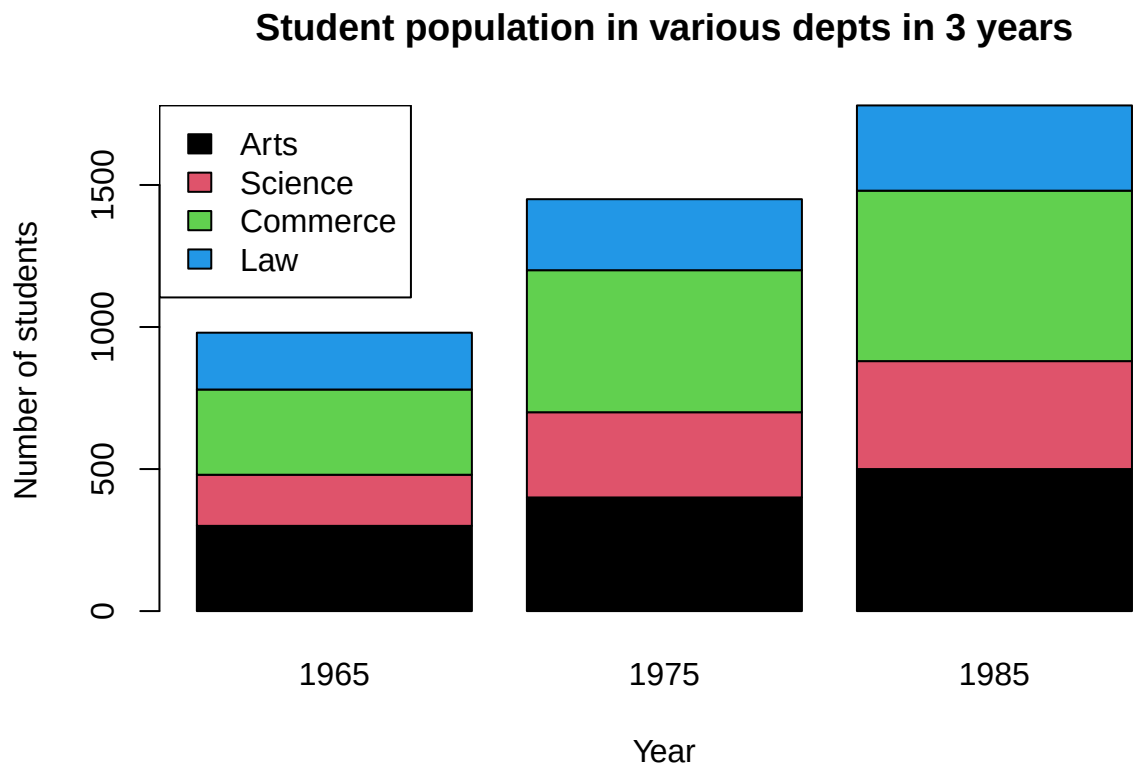



```
#b) Heatmap without column dendrogram
heatmap(as.matrix(d), main = "Heatmap", scale = "column", Colv = NA,
↪  cexCol = 0.9)
```

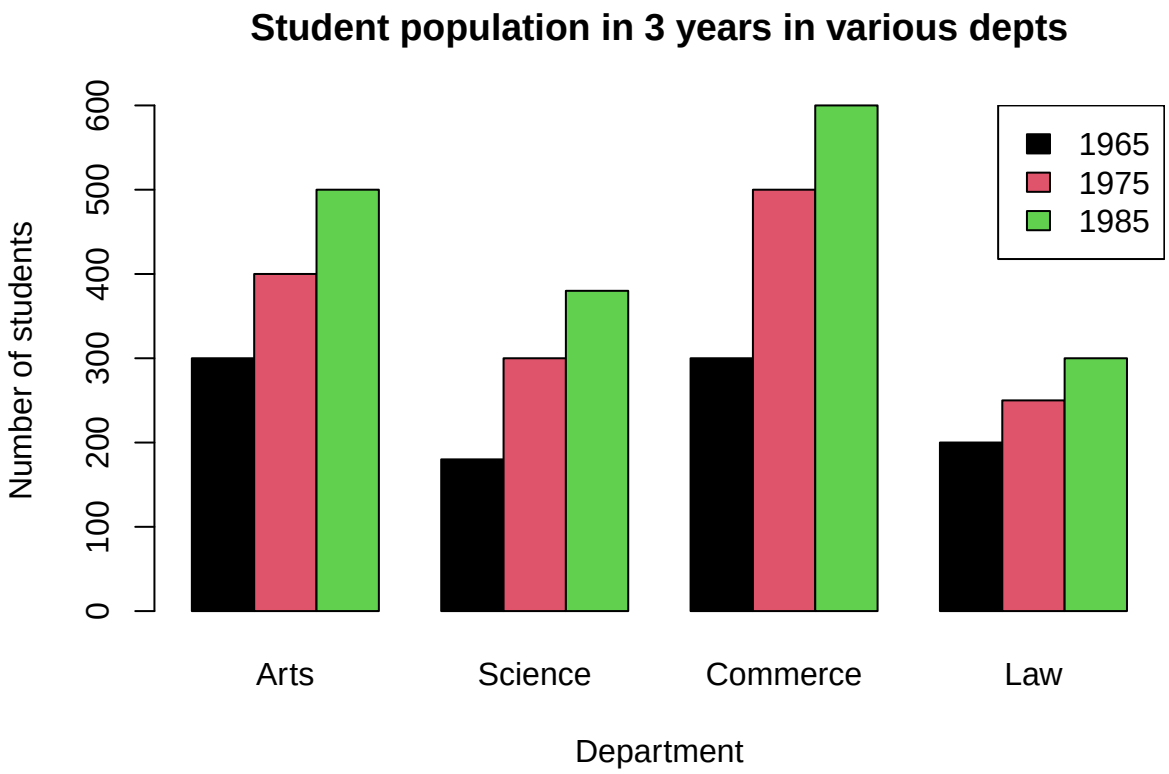

Heatmap
55
53
92
75
86
143
115
150
124
135
146
133
117
121
137
132
123
103
1
50
38
26
30
3
9
24
12
45
32
34
15
100
68
85
82
60
107
61
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width

Question 5

```r
years = c(1965, 1975, 1985)
depts = c("Arts", "Science", "Commerce","Law")
arts = c(300, 400, 500)
sci = c(180, 300, 380)
com = c(300, 500, 600)
law = c(200, 250, 300)
df = data.frame(arts, sci, com, law)
#Subdivided bar diagram
barplot(t(df), beside = F , names.arg = years, col = 1:4, xlab =
 ↪  "Year", ylab = "Number of students", main = "Student population
 ↪  in various depts in 3 years")
legend("topleft" , fill = 1 : 4, legend = depts)
```




```
#Multiple bar diagram
barplot(as.matrix(df), beside = T, names.arg = depts, col = 1:3,
 ↪  xlab = "Department", ylab = "Number of students", main =
 ↪  "Student population in 3 years in various depts")
legend("topright" , fill = 1 : 3, legend = years)
```


Student population in 3 years in various depts
Number of students
0
100
200
300
400
500
600
1965
1975
1985
Arts
Science
Commerce
Law
Department

Question 6

```r
#a)
name = c("Bhargav Singh", "Sayali Shah", "John Pinto", "Anaya
↪  Joshi", "Arnav Singhal")
position = c("Head", "Head", "Manager", "Network architect", "Senior
↪  Software Developer")
dept = c("Finance", "Human Resources", "Marketing", "IT", "R&D")
salary = c(2000000, 2200000, 1900000, 1650000, 1850000)
df = data.frame(name, position, dept, salary)
df
```

```
##            name                   position            dept  salary
## 1 Bhargav Singh                       Head         Finance 2000000
## 2   Sayali Shah                       Head Human Resources 2200000
## 3    John Pinto                    Manager       Marketing 1900000
## 4   Anaya Joshi          Network architect              IT 1650000
## 5 Arnav Singhal Senior Software Developer             R&D 1850000
```

```r
summary(df)
```

```
##      name              position             dept                salary
##  Length:5           Length:5           Length:5           Min.   :1650000
##  Class :character   Class :character   Class :character   1st Qu.:1850000
##  Mode  :character   Mode  :character   Mode  :character   Median :1900000
##                                                           Mean   :1920000
##                                                           3rd Qu.:2000000
##                                                           Max.   :2200000
```

```r
#b)
a = array(seq(52, 80, 2), dim = c(5, 3))
a
```

```
##      [,1] [,2] [,3]
## [1,]   52   62   72
## [2,]   54   64   74
## [3,]   56   66   76
## [4,]   58   68   78
## [5,]   60   70   80
```

```r
#c)
v = c(1, 2, 3, 3, 4, NA, 3, 2, 4, 5, NA, 5)
levels(factor(v))
```

```
## [1] "1" "2" "3" "4" "5"
```

```r
#d)
l = list("Probability", 3.25, 9L, "STATISTICS", c(1, 2, "PS"), 1:3,
↪  T, F)
l
```

```
## [[1]]
## [1] "Probability"
##
## [[2]]
## [1] 3.25
##
```

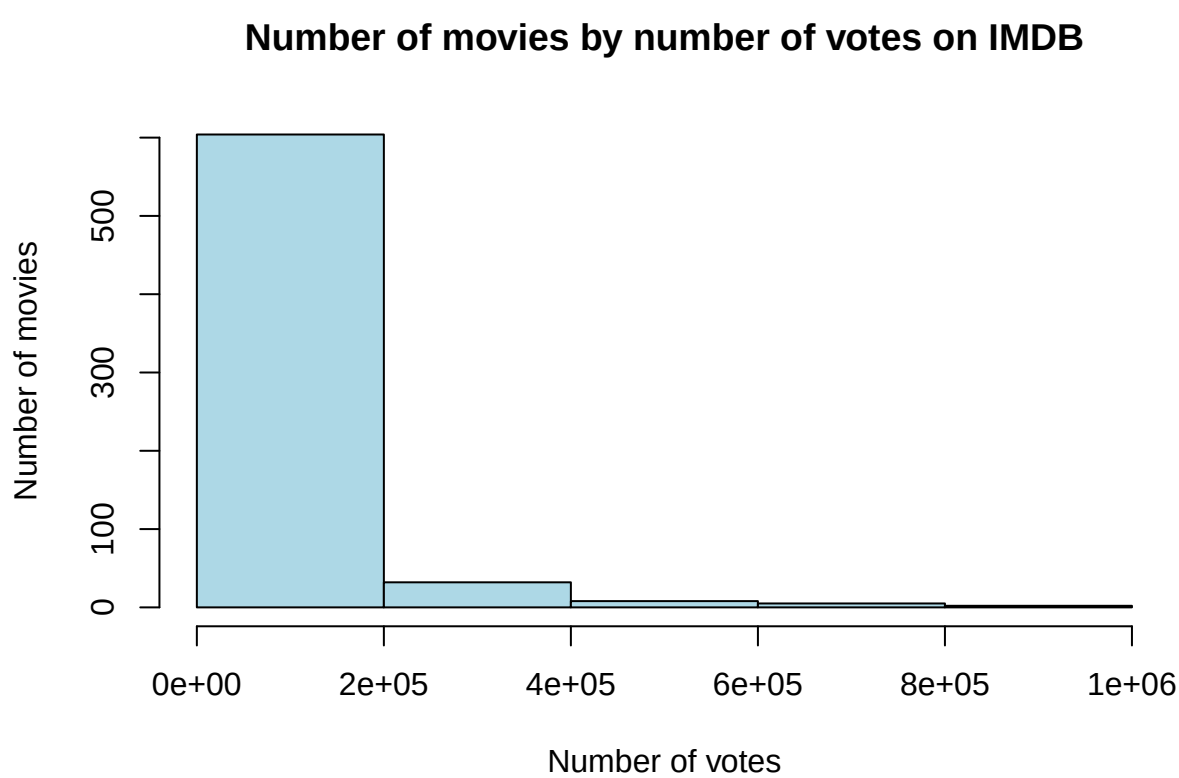
```
## [[3]]
## [1] 9
##
## [[4]]
## [1] "STATISTICS"
##
## [[5]]
## [1] "1"  "2"  "PS"
##
## [[6]]
## [1] 1 2 3
##
## [[7]]
## [1] TRUE
##
## [[8]]
## [1] FALSE
```

```r
#e)
s = sample(-50:50, 10)
s
```

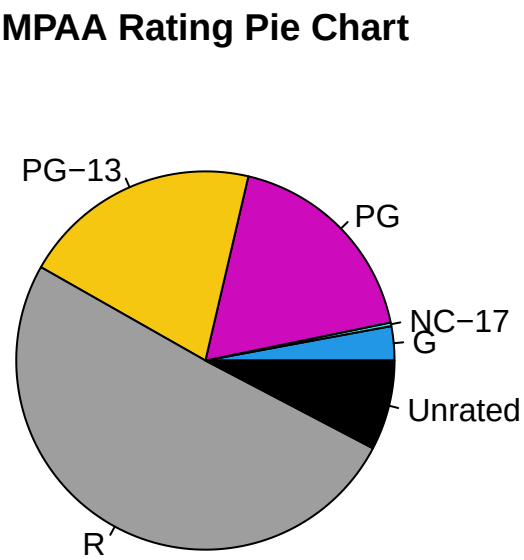
```
##  [1]  27 -33   9 -18  19  18  31  42  41 -31
```

Question 7

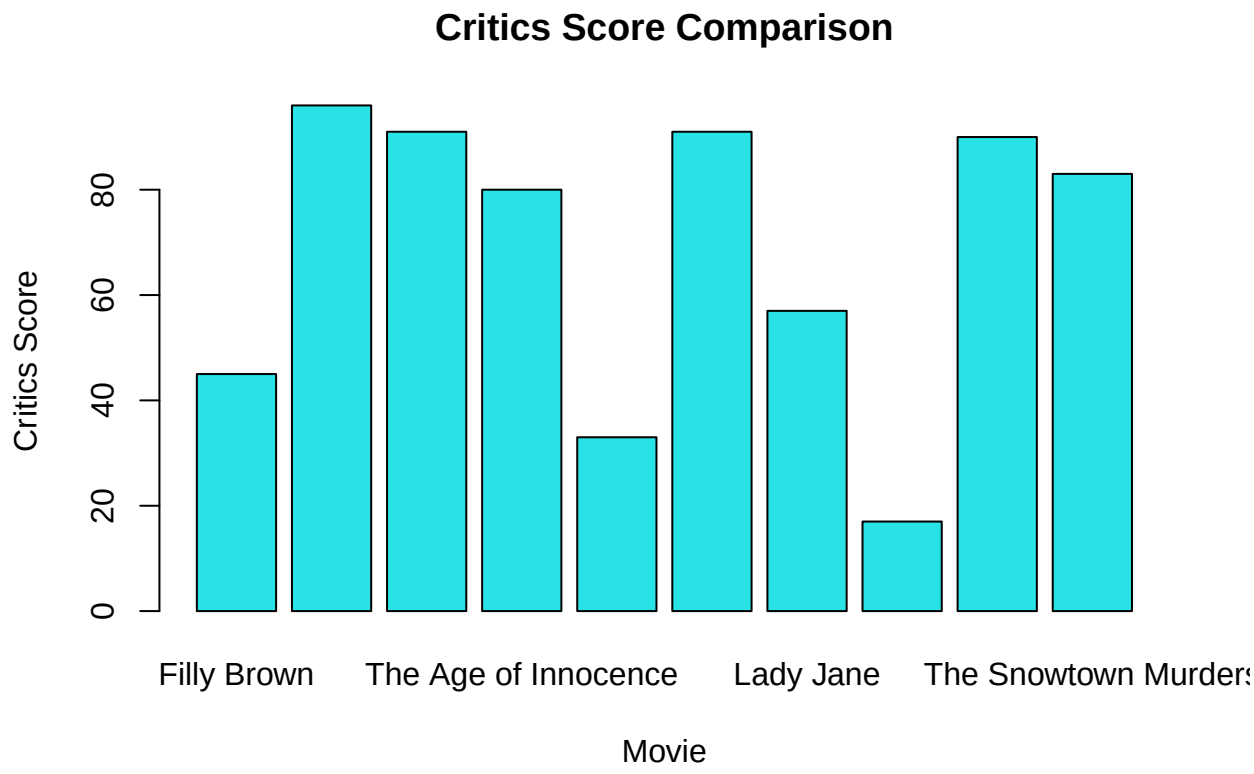
```r
#Q7
moviesData = read.csv(file =
 ↪  "C:/Users/singh/Documents/moviesData.csv", header = TRUE,
 ↪  stringsAsFactors = FALSE, na.strings = "NA")
#a)
hist(moviesData$imdb_num_votes, xlab = "Number of votes", ylab =
 ↪  "Number of movies", main = "Number of movies by number of votes
 ↪  on IMDB", col = "lightblue", breaks = 5)
```



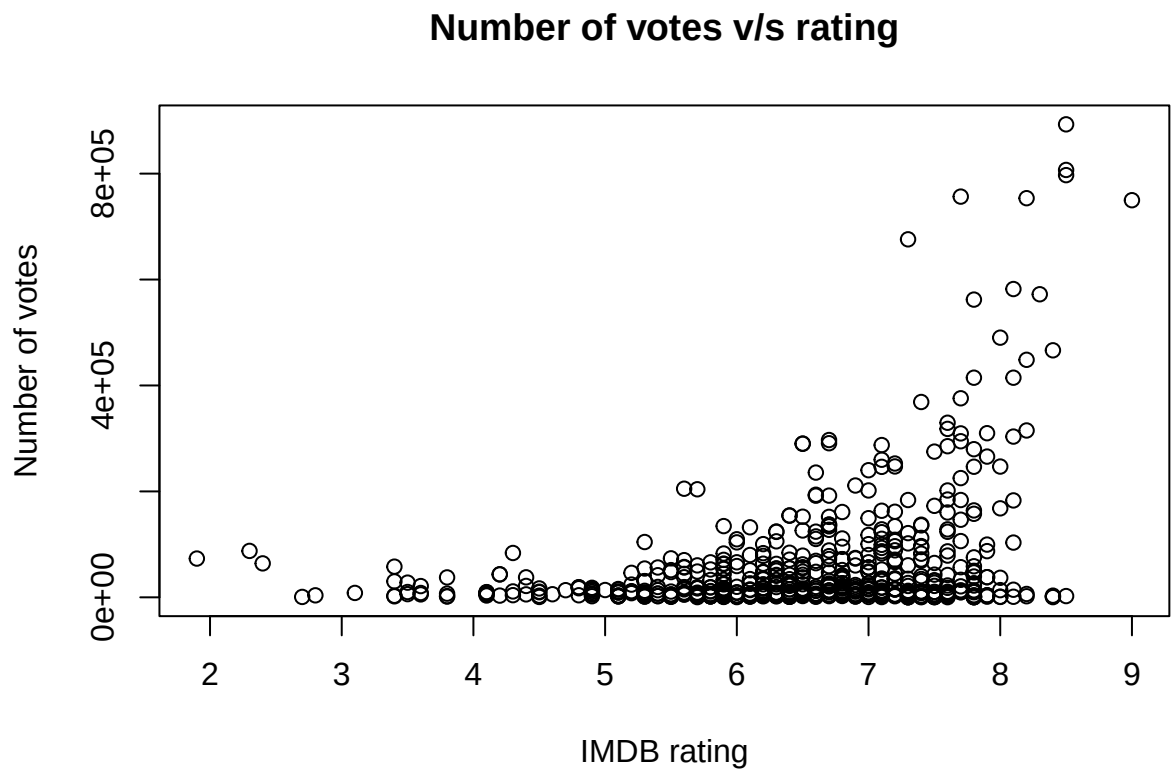
```r
#b)
pie(table(moviesData$mpaa_rating), col = 4:9, main = "MPAA Rating
↪  Pie Chart")
```



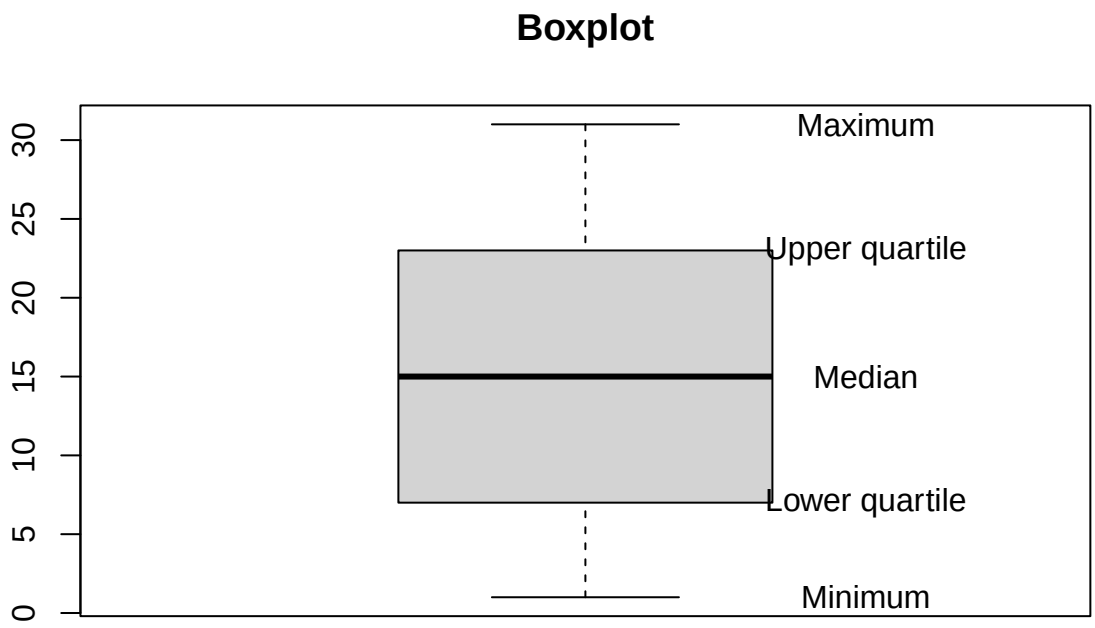

```r
#c)
m = matrix(moviesData$critics_score[1:10])
barplot(t(m), xlab = "Movie", ylab = "Critics Score", main =
↪  "Critics Score Comparison", col = 5, names.arg =
↪  moviesData$title[1:10])
```




```r
#d)
plot(moviesData$imdb_rating, moviesData$imdb_num_votes, main =
↪  "Number of votes v/s rating", xlab = "IMDB rating", ylab =
↪  "Number of votes")
```




```
#e)
x = moviesData$dvd_rel_day
f = fivenum(x)
boxplot(x, main = "Boxplot")
text(rep(1.3, 5), f, labels = c("Minimum", "Lower quartile",
 ↪  "Median", "Upper quartile", "Maximum"))
```


Boxplot
0
5
10
15
20
25
30
Maximum
Upper quartile
Median
Lower quartile
Minimum

Question 8

```r
data(CO2)
up = CO2$uptake
#a)
sqrt(var(up)) #Standard deviation = 10.81441
```

```
## [1] 10.81441
```

```r
#b)
qd = as.numeric(quantile(up, 0.75) - quantile(up, 0.25))
qd #Quartile deviation = 19.225
```

```
## [1] 19.225
```

```r
#c)
r = max(up) - min(up)
r #Range = 37.8
```

```
## [1] 37.8
```

```r
#d)
t = table(up)
m = names(t)[which(t > 1 & t == max(t))]
m #Mode = [1] "17.9" "32.4"
```

```
## [1] "17.9" "32.4"
```

```r
#e)
coeff = r / (max(up) + min(up))
coeff #Coefficient of range = 0.7105263
```

```
## [1] 0.7105263
```